VF Asia Footwear Laboratory Testing Manual

Test Method for Centrifuge Test	**Test method # VF- G09** **REF # : TM-R-006** **Page 1 of 2**
Date Issued : September 5, 2014 **Revision # :**	**Created by : Nila Babia Erazo / Chris Adam**

PURPOSE AND SCOPE: To determine the waterproofness of waterproof footwear with the use of The W.L. Gore™ Centrifuge apparatus by spinning Gore-Tex® bootie constructed footwear containing water at 231 to 234 rpm for a predetermined length of time. The areas of the footwear tested using this method include bootie seam integrity.

REFERENCE: None

EQUIPMENT AND MATERIAL:

1. W. L. Gore™ Centrifuge Machine
2. Pocket knife or equivalent
3. Timer
4. Dry footwear with Gore-Tex bootie.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0 ˚C Temperature and 55 ±10 % Relative Humidity.

2. Remove all hangtags, laces, inserts, and tissue from dry footwear.

PROCEDURE:

1. Ensure inside of Centrifuge machine, door and glass are dry.
2. Fill footwear with water to within 19mm from the lowest waterproof bootie point (footwear must be the same size and width)
3. Place footwear on the same level with one facing forward and the other facing backwards. There are two levels in the machine and each level is independent of the other allowing a different style#, size, or width to run at the same time.
4. Set centrifuge speed @ 231 to 234 rpm.

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5. Set timer.
6. Close cage/flexi-glass window, and press start button.
7. Visibly monitor for leaks (drops of water) against the flexi-glass window. If leaks are visible push the red stop button to stop the machine.
8. Remove leaking footwear, mark wet area with a silver ink pen, empty water from footwear and dry footwear on dryer machine. Record time the failure occurs and where the shoe failed.
9. After remaining footwear has reached predetermined number of minutes, stop test.
10. Remove footwear, empty water from footwear and dry on dryer machine.
11. As required perform destructive test or dismantling footwear to determine cause of leak.

REPORT:

1. Report the material description.
2. Applied pressure, test duration.
3. A pass/fail value.
4. Any deviations from this procedure.